HYDROGEN PEROXIDE (>60% SOLUTION IN WATER) **ICSC: 0164 (May 2018)**

Hydroperoxide
Hydrogen dioxide
Dihydrogen dioxide

CAS #: 7722-84-1
UN #: 2015
EC Number: 231-765-0

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Not combustible. The substance may ignite combustible materials. Risk of fire and explosion on contact with heat or metal catalysts.	NO contact with hot surfaces. NO contact with incompatible materials: See Chemical Dangers	In case of fire in the surroundings, use appropriate extinguishing media. In case of fire: keep drums, etc., cool by spraying with water.

PREVENT GENERATION OF MISTS! AVOID ALL CONTACT! IN ALL CASES CONSULT A DOCTOR!

	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Sore throat. Cough. Dizziness. Headache. Nausea. Shortness of breath.	Use ventilation, local exhaust or breathing protection.	Fresh air, rest. Half-upright position. Refer immediately for medical attention.
Skin	MAY BE ABSORBED! Skin discoloration. Swelling. Redness. Pain. Skin burns.	Protective gloves. Protective clothing.	First rinse with plenty of water for at least 15 minutes, then remove contaminated clothes and rinse again. Refer immediately for medical attention .
Eyes	Redness. Pain. Blurred vision. Corneal damage. Burns.	Wear safety goggles or face shield.	First rinse with plenty of water for several minutes (remove contact lenses if easily possible), then refer for medical attention.
Ingestion	Aspiration hazard! Sore throat. Abdominal pain. Abdominal distension. Shock or collapse.	Do not eat, drink, or smoke during work.	Rinse mouth. Do NOT induce vomiting. Refer immediately for medical attention.

SPILLAGE DISPOSAL

Consult an expert! Personal protection: chemical protection suit including self-contained breathing apparatus. Ventilation. Do NOT let this chemical enter the environment. Absorb liquid in sand or inert absorbent. Do NOT absorb in saw-dust or other combustible absorbents. Carefully collect remainder. Store and dispose of according to local regulations.

STORAGE

Store in an area without drain or sewer access. Separated from food and feedstuffs. See Chemical Dangers. Cool. Keep in the dark. Store in vented containers. Store only if stabilized.

PACKAGING

Special material.

CLASSIFICATION & LABELLING

According to UN GHS Criteria

DANGER

May cause fire or explosion; strong oxidizer
Harmful if swallowed
Toxic if inhaled
Causes severe skin burns and eye damage
May cause respiratory irritation
May cause damage to upper respiratory tract and lungs through prolonged or repeated exposure
Toxic to aquatic life

Transportation
UN Classification
UN Hazard Class: 5.1; UN Subsidiary Risks: 8; UN Pack Group: I

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PHYSICAL & CHEMICAL INFORMATION

Physical State; Appearance
COLOURLESS LIQUID.

Physical dangers

Chemical dangers
Decomposes under the influence of light. Decomposes on warming. This produces oxygen. This increases fire hazard. The substance is a strong oxidant. It reacts violently with combustible and reducing materials. This generates fire and explosion hazard particularly in the presence of metals. Attacks many organic substances such as textiles and paper.

Formula: H_2O_2
Molecular mass: 34.0
Boiling point: 141°C (90%)
Boiling point: 125°C (70%)
Melting point: -11°C (90%), -39°C (70%)
Relative density (water = 1): 1.4 (90%)
Relative density (water = 1): 1.3 (70%)
Solubility in water: miscible
Vapour pressure, kPa at 20°C: 0.2 (90%)
Vapour pressure, kPa at 20°C: 0.1 (70%)
Relative vapour density (air = 1): 1
Relative density of the vapour/air-mixture at 20°C (air = 1): 1.0
Octanol/water partition coefficient as log Pow: -1.36

EXPOSURE & HEALTH EFFECTS

Routes of exposure
The substance can be absorbed into the body by inhalation of its vapour, by ingestion and through the skin.

Effects of short-term exposure
The substance is corrosive to the eyes, skin and respiratory tract. Corrosive on ingestion. The vapour is severely irritating to the respiratory tract. Ingestion may cause strong foam formation with risk of asphyxiation and aspiration. Exposure to this substance may produce oxygen bubbles (embolism) in the blood, resulting in shock.

Inhalation risk
A harmful contamination of the air can be reached rather quickly on evaporation of this substance at 20°C.

Effects of long-term or repeated exposure
Repeated or chronic inhalation of the vapour may cause chronic inflammation of the upper respiratory tract. Lungs may be affected by repeated or prolongated exposure. The substance may have effects on the hair. This may result in bleaching.

OCCUPATIONAL EXPOSURE LIMITS

TLV: 1 ppm as TWA; A3 (confirmed animal carcinogen with unknown relevance to humans).
MAK: 0.71 mg/m^3, 0.5 ppm; peak limitation category: I(1); carcinogen category: 4; pregnancy risk group: C

ENVIRONMENT

The substance is toxic to aquatic organisms.

NOTES

Rinse contaminated clothing with plenty of water because of fire hazard.
Other UN numbers: 2014 (hydrogen peroxide, aqueous solution 20-60%): hazard class 5.1, subsidiary hazard 8, pack group II; 2984 (hydrogen peroxide, aqueous solution 8-20%): hazard class 5.1, pack group III.

ADDITIONAL INFORMATION

EC Classification